

# Strategic coexistence theory for evolutionary games, another case study

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So far, we have focused on simple deterministic  $2 \times 2$  games. However, many ecological and evolutionary dynamics are intrinsically stochastic. Recently, Wang et al. (2025) showed that environmental fluctuations can promote coexistence in games that would otherwise lead to competitive exclusion in a deterministic environment. From the perspective of SCT, this suggests that stochasticity can stabilize competition by promoting the recovery of strategies when rare. Interestingly, Wang et al. (2025) also showed that stochasticity can generate bistable coexistence and multiple internal equilibria. Thus, although stochasticity can stabilize competition near the boundaries, it can also generate destabilizing frequency dependence in the interior. Here, we use SCT to separate these effects and clarify how stochasticity can both stabilize and destabilize strategic competition.

Following Wang et al. (2025), we consider a  $2 \times 2$  game in the presence of environmental noise. The stochastic payoff matrix is given by

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix} + \frac{\zeta}{\sqrt{s}} \begin{pmatrix} \sigma_R & \sigma_S \\ \sigma_T & \sigma_P \end{pmatrix}, \quad (1)$$

where the first matrix represents deterministic payoffs,  $\zeta$  represents a noise term with zero mean and unit variance, and the second matrix determines the intensity of environmental stochasticity. Assuming weak selection  $s \ll 1$ , the evolution of two competing strategies in a large population can be described by the following equation:

$$\frac{dx}{dt} = sx(1-x) \left( \pi_C - \pi_D + \left( \frac{1}{2} - x \right) (\sigma_C - \sigma_D)^2 \right), \quad (2)$$

where  $\pi_C = Rx + S(1-x)$  and  $\pi_D = Tx + P(1-x)$  and  $\sigma_C = \sigma_R x + \sigma_S(1-x)$  and  $\sigma_D = \sigma_T x + \sigma_P(1-x)$ .

Then, an extended SCT decomposition shows that niche overlap and fitness difference can be written as:

$$\rho = \exp \left( \frac{R + P - S - T}{2} - \frac{(\sigma_R - \sigma_T)^2 + (\sigma_S - \sigma_P)^2}{4} \right) \quad (3)$$

$$\frac{\kappa_D}{\kappa_C} = \exp \left( \frac{T + P - R - S}{2} + \frac{(\sigma_R - \sigma_T)^2 - (\sigma_S - \sigma_P)^2}{4} \right) \quad (4)$$

31 Writing  $\hat{\rho}$  and  $\hat{\kappa}_D/\hat{\kappa}_C$  to denote niche overlap and fitness difference in the absence of  
 32 stochasticity, we have:

$$\rho = \hat{\rho} \exp \left( -\frac{(\sigma_R - \sigma_T)^2 + (\sigma_S - \sigma_P)^2}{4} \right) \quad (5)$$

$$\frac{\kappa_D}{\kappa_C} = \frac{\hat{\kappa}_D}{\hat{\kappa}_C} \exp \left( \frac{(\sigma_R - \sigma_T)^2 - (\sigma_S - \sigma_P)^2}{4} \right). \quad (6)$$

33 Therefore, stochasticity always reduces niche overlap, thereby increasing niche dif-  
 34 ference and stabilizing competition. In contrast, it can have opposing effects on

## 35 Supp

36 Consider a competition of two strategies under stochastic payoffs:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix} + \frac{\zeta}{\sqrt{s}} \begin{pmatrix} \sigma_R & \sigma_S \\ \sigma_T & \sigma_P \end{pmatrix}, \quad (7)$$

37 Assuming weak selection  $s \ll 1$ , the evolution of two competing strategies in a large  
 38 population can be described by the following equation:

$$\frac{dx}{dt} = sx(1-x) \left( \pi_C - \pi_D + \left( \frac{1}{2} - x \right) (\sigma_C - \sigma_D)^2 \right), \quad (8)$$

39 To simplify, we write

$$\frac{dx}{dt} = sx(1-x)G(x), \quad (9)$$

40 where

$$G(x) = \left( \pi_C - \pi_D + \left( \frac{1}{2} - x \right) (\sigma_C - \sigma_D)^2 \right). \quad (10)$$

41 Thus, the sign of  $G(x)$  at boundary conditions determines whether a strategy can  
 42 invade when the competing strategy is at equilibrium.

43 Note that we cannot immediately apply the original SCT presented in the main  
 44 text here because  $G(x)$  does not uniquely separate into two payoff measures. In-  
 45 stead, we can still utilize the invasion growth rate when rare and follow the same  
 46 procedure. In particular, the mutual invasion condition at monomorphic boundary  
 47 can be written as:

$$0 < G(0) \quad (11)$$

$$0 < -G(1) \quad (12)$$

48 Exponentiating both, we can combine them into a single inequality:

$$\exp(G(1)) < 1 < \exp(G(0)) \quad (13)$$

49 Dividing both sides by  $\exp((G(1) + G(0))/2)$ , we get:

$$\exp\left(\frac{G(1) - G(0)}{2}\right) < \exp\left(-\frac{G(1) + G(0)}{2}\right) < \exp\left(\frac{G(0) - G(1)}{2}\right), \quad (14)$$

50 which gives us expressions for niche overlap and fitness difference:

$$\rho = \exp\left(\frac{G(1) - G(0)}{2}\right) \quad (15)$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(-\frac{G(1) + G(0)}{2}\right) \quad (16)$$

51 Then, by substituting expressions for  $G(0)$  and  $G(1)$ , we obtain following expressions  
52 for niche and fitness difference:

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2} - \frac{(\sigma_R - \sigma_T)^2 + (\sigma_S - \sigma_P)^2}{4}\right) \quad (17)$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(\frac{T + P - R - S}{2} + \frac{(\sigma_R - \sigma_T)^2 - (\sigma_S - \sigma_P)^2}{4}\right) \quad (18)$$

## 53 References

54 Wang, G., Q. Su, L. Wang, and J. B. Plotkin (2025). Evolution of social behaviors  
55 in noisy environments. *arXiv*.